KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) State the problems of a stop-n-wait type reliable data transfer protocol. Design a reliable, pipelined data transfer protocol that uses no buffer for out of order packets. (15)
   b) Briefly describe the services provided by TCP. (09)
   c) Show the format of HTTP request line. Mention at least two request type used by HTTP version 1.1 and explain. (06)
   d) Explain common FTP commands. (05)

2. a) Consider transferring an enormous file of L bytes from host A to host B. What is the maximum value of L such that TCP sequence numbers are not exhausted? Assume an MSS of 1460 bytes. (05)
   b) The TCP of a host receives five segments at the time shown (HH:MM:SS.mmm) below. If it has no data to send, show the acknowledgements sent by the host using line diagram. (09)
   (i) at 0:0:0.000 (ii) at 0:0:0.027 (iii) at 0:0:0.400 (iv) at 0:0:1.200 (v) at 0:0:1.208
   c) Show a sample SMTP interaction between SMTP client and server to transfer the message "Good morning my friend" from alice@xxx.com to bob@yyy.com. (11)
   d) In TCP, what problems are solved by the following timers and how? (10)
   (i) Persistence timer (ii) Time-waited timer

3. a) Suppose a DNS client is looking for the IP address of www.kuet.ac.bd. It already sent a query message to the DNS server. Show the response message sent by the server. Use necessary DNS compression. Assume the IP address is 128.175.13.63. (10)
   b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (09)
   c) Since UDP is unreliable, why is it used at transport layer? (05)
   d) What do you mean by digital signature? How does RSA provides digital signature mechanism? (11)

4. a) (15)

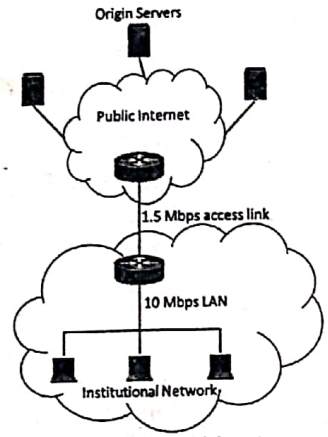


Figure 4(a)

Consider Figure 4(a), for which there is an institutional network connected to the internet. Let the average object size is 900 kbit and the average request rate from institutional browsers to the origin servers is 1.5 request per second. Also suppose that the amount of time it takes from when the router on the internet side of the access link forwards an HTTP request until it receives the response in two seconds on average. Model the total average response time as a sum of the average access delay and the average internet delay. For average access delay, use $\Delta/(1-\Delta\beta)$, where Δ is the time required to send an object over the access link and β is the arrival rate of objects to the access link.

i) Find the total average response time.
ii) Now suppose a web cache is installed in the institutional LAN. Let the hit rate is 0.4. Find the total response time.

b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)

c) List down the steps involved in SSL handshaking. Also present a pictorial view of packets transmission. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) How does header checksum can be calculated in IPv4 datagram? Why does it cover only data portion? Explain with example. (12)

b) Define Computer Network. What are the major functionalities of Computer Network? (08)

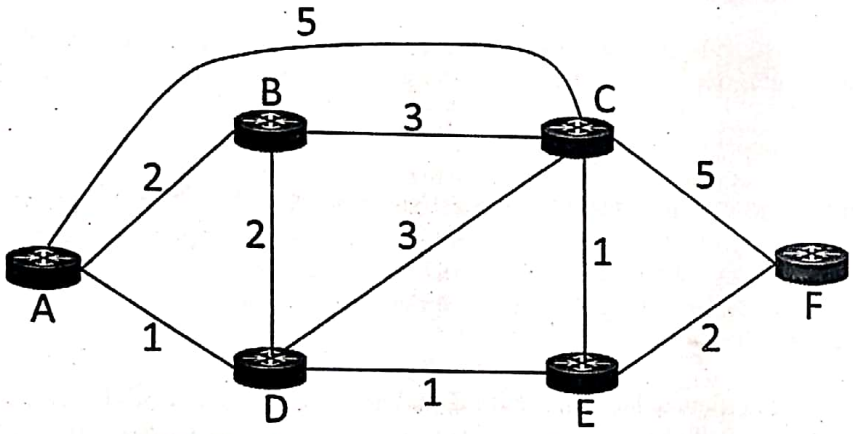
c) Apply Fragmentation on following information for IPv4 Datagram. (15)

i) Original Datagram : 5000 bytes, Header : 20 bytes.

ii) MTU : First level 2000 bytes and Second level 800 bytes.

6. a) Concisely describe the error reporting messages of ICMP. (10)

b) Determine the routing table/forwarding table at the router B from the following topology using a Link-State Routing Algorithm. (17)



c) "Switch is more intelligent than hub" – justify the statement. (08)

7. a) A router inside an organization receives a packet with the destination 190.240.33.91. How does it find the subnetwork address to route the packet? Assume the subnet mask is /19. (05)

b) Write a short note on Address Resolution Protocol (ARP). (09)

c) What is count to infinity problem? Use a suitable example to show that the poisoned reverse technique can solve the problem. (12)

d) Why do we need Network Address Translation (NAT)? Briefly explain how NAT works. (09)

8. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R. (12)

b) How does Slotted ALOHA random access MAC protocol work? Show the efficiency of it. (13)

c) Write down the services provided by the data link layer. (10)